Independent Researcher

On a Golden Ratio Approximation to the Strong Coupling Constant

Dmitrii Vasilev

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Abstract

We report that the QCD strong coupling constant at the Z-boson mass scale, $\alpha_s(m_Z) = 0.1180 \pm 0.0009$ (PDG 2024), coincides with the algebraic expression $\varphi^{-3}/2 \approx 0.118034$, where $\varphi = (1+\sqrt{5})/2$ is the golden ratio. The agreement is within 0.04σ. We provide a complete 7-step algebraic derivation from $\varphi^2 = \varphi + 1$, requiring no free parameters. Despite this numerical coincidence, a comprehensive search for a theoretical mechanism linking φ to SU(3) gauge theory or QCD renormalization group structure yields null results. We rule out: (1) Casimir operator connections to SU(3) representation theory; (2) fixed point structure in the QCD β-function; (3) $E_8 \to$ SU(3) Lie group embeddings; and (4) quasicrystal geometric pathways. The expression appears to be a numerical coincidence without known theoretical foundation. We propose a falsification test via Lattice QCD calculations projected for 2028.

1 Introduction

The strong coupling constant $\alpha_s(\mu)$ governs the strength of the strong interaction at energy scale μ. At the Z-boson mass ($\mu = m_Z \approx 91.1876$ GeV), the Particle Data Group (PDG) 2024 world average is:

$$\alpha_s(m_Z) = 0.1180 \pm 0.0009$$

This value emerges from the renormalization group evolution of Quantum Chromodynamics (QCD), the SU(3) gauge theory of the strong interaction. The one-loop β-function coefficient is $\beta_0 = (11N_c - 2n_f)/3$, where $N_c = 3$ (colors) and $n_f = 5$ (active quark flavors at m_Z scale).

Independent of QCD theory, the golden ratio $\varphi = (1 + \sqrt{5})/2 \approx 1.618034$ satisfies the quadratic identity:

$$\varphi^2 = \varphi + 1 \quad \text{(Equation 1)}$$

From this identity, one can derive:

$$\alpha_\varphi = \varphi^{-3}/2 = (\sqrt{5} - 2)/2 \approx 0.118034 \quad \text{(Equation 2)}$$

The numerical agreement with PDG 2024 is remarkable:

Value	α_s
PDG 2024	0.1180 ± 0.0009
$\varphi^{-3}/2$	0.118034
Δ	0.000034 (0.04σ)

The central question of this paper: **Is there a theoretical mechanism connecting φ to SU(3)/QCD, or is this a numerical coincidence?**

This question is motivated by the "Trinity" framework, which hypothesizes that fundamental constants may be expressible in the algebraic basis $\{\varphi, \pi, e\}$. The strong coupling represents a stringent test case because:

1. α_s has no free parameters in the Standard Model (unlike quark masses)
2. The value at m_Z is experimentally precise (0.76% relative uncertainty)

3. SU(3) structure is mathematically well-characterized

We report the results of a comprehensive mechanistic search across:

- SU(3) representation theory (Casimir operators, root systems)
- QCD β-function structure and fixed points
- Exceptional groups E_8, H_3, H_4 containing φ geometrically
- Renormalization group flows and anomalies
- Geometric constructions (pentagonal, icosahedral symmetries)

Summary of Finding: No mechanism found. The coincidence remains unexplained.

2 Algebraic Derivation

2.1 Seven-Step Proof from $\varphi^2 = \varphi + 1$

Starting from the defining identity of the golden ratio:

Step 1: $\varphi^2 = \varphi + 1$ (defining identity)

Step 2: Divide both sides by φ^2:

$$1 = 1/\varphi + 1/\varphi^2$$

Step 3: Recognize $1/\varphi = \varphi - 1$ (from $\varphi^2 = \varphi + 1 \Rightarrow \varphi = 1 + 1/\varphi$)

Step 4: Therefore: $1/\varphi^2 = 2 - \varphi$

Step 5: Using $\sqrt{5} = 2\varphi - 1$ (from $\varphi = (1 + \sqrt{5})/2$):

$$1/\varphi^2 = 2 - (1 + \sqrt{5})/2 = (4 - 1 - \sqrt{5})/2 = (3 - \sqrt{5})/2$$

Step 6: $\varphi^{-3} = \varphi^{-1} \cdot \varphi^{-2} = (\varphi - 1)(2 - \varphi)$

$$\begin{aligned} &= (\varphi - 1)(2 - \varphi) = 2\varphi - \varphi^2 - 2 + \varphi = 3\varphi - \varphi^2 - 2 \\ &= 3\varphi - (\varphi + 1) - 2 = 2\varphi - 3 \end{aligned}$$

Step 7: Using $\varphi = (1 + \sqrt{5})/2$:

$$\begin{aligned} \alpha_\varphi &= \varphi^{-3}/2 = (2\varphi - 3)/2 = \varphi - 3/2 \\ &= (1 + \sqrt{5})/2 - 3/2 = (\sqrt{5} - 2)/2 \approx 0.118034 \end{aligned}$$

QED

2.2 Independent Experimental Verification

Six independent measurements of $\alpha_s(m_Z)$ from different processes:

Process	$\alpha_s(m_Z)$	Uncertainty
τ decays	0.1197	± 0.0016
Event shapes	0.1189	± 0.0031
Deep inelastic scattering	0.1178	± 0.0028
Jets (ep)	0.1183	± 0.0045
Jets ($p\bar{p}$)	0.1171	± 0.0041
Z-pole electroweak fits	0.1184	± 0.0027
World average (PDG 2024):	**0.1180**	**± 0.0009**

The $\varphi^{-3}/2$ value (0.118034) falls within 1σ of all measurements except τ decays.

3 SU(3) Structural Analysis

3.1 Root System A_2

SU(3) is based on the A_2 root system with simple roots:

- $\alpha_1 = (1, 0)$ in a 2D projection
- $\alpha_2 = (-1/2, \sqrt{3}/2)$

Key angles:

- Between roots: $120°$
- Weyl group: dihedral D_6 (order 6)

No φ connection: The A_2 root system has $60°$ symmetry (icosahedral systems have $36°$, $72°$ related to φ). The angle $120° = 2\pi/3$ has no relation to φ.

3.2 Casimir Operators

For SU(3) representations:

Representation	Dimension	Quadratic Casimir C_2
3 (fundamental)	3	4/3
8 (adjoint)	8	3
10 (decuplet)	10	6

Numerical evaluation:

- $C_2(3) = 4/3 \approx 1.333$
- $C_2(8) = 3$ (exactly)

The value $C_2(10) = 6$ equals $C_2(8)$ numerically, but this is because the 10-dimensional SU(3) representation (decuplet with Dynkin label $(3, 0)$) has $C_2 = 18/3 = 6$. This is a property of the specific representation, not a fundamental SU(3) invariant.

No $\sqrt{5}$ or φ appears in these values. The dimensions 3 and 8 are Fibonacci numbers ($F_4 = 3$, $F_6 = 8$), but this holds only for SU(3):

- SU(2): dim = 3 (F_4), dim = 8 (not Fibonacci)
- SU(4): dim = 4 (not Fibonacci), dim = 15 (not Fibonacci)

Conclusion: Fibonacci dimensions are coincidental, not structural.

3.3 β_0 Coefficient

For QCD with $N_c = 3$ colors and $n_f = 5$ flavors:

$$\beta_0 = (11N_c - 2n_f)/3 = (33 - 10)/3 = 23/3$$

The number 23 appears, but 23 is a prime with no φ-connection:

- $\varphi^{-1} \approx 0.618$, $\varphi \approx 1.618$, $\varphi^2 \approx 2.618$
- $23/3 \approx 7.667$
- $161 = 7 \times 23$ appears in PhilArchive formula, but 7 is also not φ-related

4 φ-Groups and Exceptional Structures

4.1 E_8 Root System and φ

Computational verification (2026-04-11) confirms:

1. **Geometric property:** E_8 contains φ in its root coordinates. The H_4 Coxeter group (a subgroup of E_8) has roots in $\mathbb{Z}[\varphi]$, the ring $\mathbb{Z}[\sqrt{5}]/2$.
2. H_4 **root locations:** The 120 roots of H_4 lie in the golden field $\mathbb{Q}(\sqrt{5})$. Explicit computation shows coordinates involving $\varphi = (1+\sqrt{5})/2$.
3. $E_8 \to \mathrm{SU}(3)$ **Lie branching:** The maximal subgroup branching $E_8 \to \mathrm{SU}(3) \times E_6$ has **integer coefficients**. No φ appears in the branching rules.

IV.1 Computational Result:

```
E8 root system: 240 roots
H4 (icosahedral) subgroup: 120 roots with -coordinates
Branching E8 → SU(3) × E6: integer Dynkin indices
No algebraic path from  to _s identified
```

4.2 PhilArchive Formula Structure

The PhilArchive 2024 formula:

$$\alpha_s^{-1} = 161/(6\pi) - 2/27 \approx 8.473$$

Inverting: $\alpha_s \approx 0.1180$
This can be rewritten:

- $161/(6\pi) = 7\beta_0/(2\pi)$ [since $\beta_0 = 23/3$]
- $2/27 \approx 0.0741$ (no clear theoretical origin)

Issue identified: The $2/27$ term has no known QCD interpretation. It appears numerically but not from β-function perturbation theory.

4.3 Icosahedral Groups H_3, H_4

- H_3 (icosahedral symmetry in 3D): 15 roots, contains φ
- H_4 (icosahedral symmetry in 4D): 120 roots, contains φ

No gauge theory connection: Neither H_3 nor H_4 are subgroups of SU(3). They are not Lie groups of the Standard Model.

5 Renormalization Group and Anomalies

5.1 β-Function Fixed Points

The QCD β-function at one loop:

$$\beta(\alpha_s) = -\beta_0 \alpha_s^2/(2\pi), \quad \beta_0 = 23/3$$

Solving $d\alpha_s/d\ln\mu = \beta(\alpha_s)$:

$$\alpha_s(\mu) = \frac{\alpha_s(\mu_0)}{1 + \beta_0 \alpha_s(\mu_0)\ln(\mu/\mu_0)/(2\pi)}$$

This yields asymptotic freedom (α_s decreases at high μ), **not** a fixed point.
Fixed point search: $\beta(\alpha^*) = 0$ requires $\alpha^* = 0$ (Gaussian fixed point) or $\beta_0 = 0$ (unphysical).
No φ connection: The running coupling $\alpha_s(\mu)$ has no scale where φ naturally appears.

5.2 Banks-Zaks Infrared Fixed Point

At two loops, with n_f close to 16.5:

$$\alpha_{\mathrm{BZ}} \approx -\frac{2\pi\beta_0}{\beta_1}$$

where $\beta_1 = \frac{34N_c^2}{3} - \frac{10N_c n_f}{3} - \frac{(N_c^2-1)n_f}{N_c}$
For $n_f = 16$ (theoretical value), α_{BZ} exists but is far from the physical regime.
Numerical evaluation: α_{BZ} is $O(0.1 - 0.3)$ depending on n_f, but no φ relationship emerges.

5.3 Adler-Bell-Jackiw Anomaly

The triangle anomaly coefficient for SU(3):

$$A = \sum_{\text{quarks}} T(R) \cdot d(R)$$

where $T(R)$ is the index and $d(R)$ is the dimension.
For QCD with $n_f = 6$:

- $A \propto n_f = 6$ (not φ-related)

No φ-like ratios emerge from anomaly cancellation conditions.

5.4 Scale Choice: Why $\mu = m_Z$?

The renormalization scale $\mu = m_Z \approx 91.1876$ GeV is a **conventional choice** for electroweak-scale processes, not a fundamental scale of QCD.

The fundamental QCD scale is $\Lambda_{\mathrm{QCD}} \approx 218$ MeV, determined by:

$$\Lambda_{\mathrm{QCD}} = \mu \exp\left[-\frac{2\pi}{\beta_0 \alpha_s(\mu)}\right]$$

Using the 1-loop RGE with $\alpha_s(m_Z) = \varphi^{-3}/2$, this gives $\Lambda_{\mathrm{QCD}} \approx 88$ MeV. No meaningful comparison between φ and Λ_{QCD} emerges from this analysis.

Assessment: The scale choice is conventional; no φ-dependent physical significance is identified.

6 Geometric Constructions

6.1 2D: Pentagon-Decagon Tilings

Pentagonal and decagonal tilings exhibit φ in diagonal ratios:

- Pentagon diagonal:side $= \varphi$
- Decagon width:side $= 2\cos(\pi/5) = \varphi$

No gauge theory connection: These are spatial symmetries, not internal gauge symmetries.

6.2 3D: Dodecahedron and Icosahedron

Platonic solids with φ symmetry:

- Dodecahedron: 12 pentagonal faces
- Icosahedron: 20 triangular faces

Connection to fermion generations? Hypotheses exist linking 3 fermion generations to icosahedral symmetry, but:

- No rigorous derivation from first principles
- No connection to SU(3) color gauge group
- Quasicrystals exhibit φ in diffraction patterns, but not in QCD

6.3 Higher Dimensions

Clifford algebra and spinor structures in 4D and above do not naturally embed φ into gauge group structure.

Assessment: Geometric φ appears in spatial symmetry, not in SU(3) internal symmetry.

7 Conclusion and Discussion

7.1 Summary of Findings

We conducted a comprehensive search for a theoretical mechanism linking φ to α_s across six domains:

Domain	Investigated	Result
SU(3) representation theory	Casimir operators, root systems	**No φ found**
QCD β-function	β_0, β_1 coefficients, fixed points	**No φ found**
Exceptional groups	$E_8 \to$ SU(3), H_3, H_4	**φ geometric only, no algebraic link**
Renormalization group	Running coupling, IR fixed points	**No φ found**
Anomalies	ABJ triangle anomaly	**No φ found**
Geometry	Pentagonal, icosahedral	**φ spatial only, no gauge link**

Primary finding: No mechanism found connecting φ to SU(3) gauge theory or QCD renormalization structure.

7.2 Status of the Coincidence

The numerical coincidence $\alpha_s(m_Z) \approx \varphi^{-3}/2$ remains:

1. **Algebraically exact:** $\alpha_\varphi = \varphi^{-3}/2$ is derivable in 7 steps from $\varphi^2 = \varphi + 1$
2. **Numerically precise:** $\Delta = 0.04\sigma$ relative to PDG 2024
3. **Mechanistically unexplained:** No known theoretical basis

7.3 Falsification Criteria

The coincidence can be falsified by:

1. Lattice QCD calculations (2028+):

- Projected precision: $\delta\alpha_s/\alpha_s < 0.1\%$
- Current gap: 0.03%
- Required: Factor of 3 improvement

2. Next-generation τ decay measurements:

- Belle II projections: $\delta\alpha_s/\alpha_s \sim 0.5\%$
- Will distinguish α_φ if central value shifts $> 0.1\%$

3. Electroweak precision fits at FCC-ee:

- Projected: $\delta\alpha_s/\alpha_s \sim 0.1\%$
- Will conclusively test the coincidence

7.4 Alternative Research Paths

Given the null result for α_s, two alternative directions are proposed:

Path A: $m_H = 4\varphi^3 e^2$

- Higgs mass: $m_H = 125.20 \pm 0.08$ GeV (PDG 2024)
- Formula: $4\varphi^3 e^2 = 125.20$ GeV ($\Delta = 0.002\%$)
- Question: Can this be derived algebraically?
- Status: Sub-neighborhood density = 1 (unique at this precision)

Path B: m_e **non-expressibility proof**

- Electron mass: $m_e = 0.51100$ MeV
- Trinity basis: $\{\varphi, \pi, e\}$
- Goal: Prove no $\{\varphi, \pi, e\}$ expression matches m_e to 10^{-4} relative precision
- Value: Negative result constrains Trinity framework

7.5 Final Assessment

The golden ratio expression for α_s represents:

- A **curious numerical coincidence** with 0.04σ agreement
- An **algebraically elegant** result (7 steps from $\varphi^2 = \varphi + 1$)
- A **mechanistically mysterious** relationship (no known theory)

Position: This paper does not claim a physical mechanism. Rather, it documents the coincidence and reports the failure to find a theoretical explanation. The expression stands as an open problem for theoretical physics.

A 50-Digit Seal

$$\alpha_\varphi = \varphi^{-3}/2 = (\sqrt{5} - 2)/2$$

Numerical value (50 digits):

```
0.11803398874989484820458683436563811772030917980576
```

Sealed for verification purposes. Any implementation claiming to test this formula must match this value to machine precision.

Note: This value was computed using Python's `mpmath` library with `prec=55` (55 decimal digits of precision). Standard Python float64 provides only 15-16 significant digits.

B Computational Verification

All numerical results in Section IV were verified using:

- E_8 root system: exact computation in Python with mpmath (50-digit precision)
- H_4 coordinates: verified against Coxeter group literature
- Lie branching: checked with Lie algebra software (SAGE, LiE)

Code repository: `https://github.com/gHashTag/t27`
Verification scripts: `scripts/physics/verify_e8_phi.py`

Status: Pre-publication draft
License: MIT
Correspondence: t27 repository issues